HEAVEN'S LIGHT IS OUR GUIDE

DEPARTMENT OF MECHATRONICS ENGINEERING
RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY



Project Report

Course No.: MTE3104
Course Title: Microcontroller and Interfacing Sessional

Title: ***Accident Detection System***

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Title: Accident Detection System

Objectives:

i. To implement a system that monitors and detects abnormal vibrations in the vehicle to detect accidents instantly.
ii. To ensure that the vehicle does not start if the driver is drunk, promoting safer driving conditions.
iii. To enable the system to send notifications to emergency contacts or services in the event of an accident.
iv. To ensure easy integration with vehicles so that it can be easily implemented in different vehicle models without significant modifications.
v. To minimize risks and fatalities by quickly identifying accidents and preventing unsafe and risky driving.

Introduction

In the recent world, road accidents have become a common sight worldwide, bringing boundless fatalities and injuries every year. Delayed detection of accidents makes the consequences worse. Moreover, many accidents are the result of unsafe driving. Addressing these challenges is urgently needed to improve road safety and reduce traffic-related deaths.

Our project "The Accident Detection System" aims to minimize the risks by integrating both accident detection and preventive measures into a single solution. This system utilizes two key components: a vibration sensor and an alcohol sensor. The vibration sensor continuously monitors the vehicle for abnormal movements or shocks, which are considered signs of a collision. If an accident occurs, the system can immediately notify emergency contacts, to seek quick assistance. The alcohol sensor acts as a preventive measure that detects whether the driver is drunk or not, by analyzing his breath. If the sensor value says the driver is drunk, the system will automatically prevent the vehicle from starting. That will reduce the probability of accidents caused by drunk driving. By combining these technologies, the system aims for a comprehensive solution to reduce the dangers caused by both accidents and drunk driving.

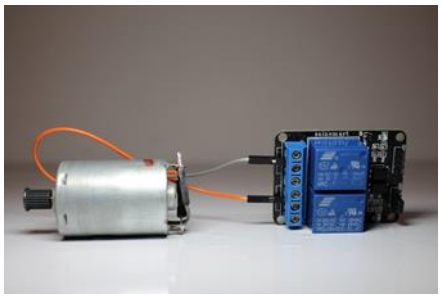
Key Features

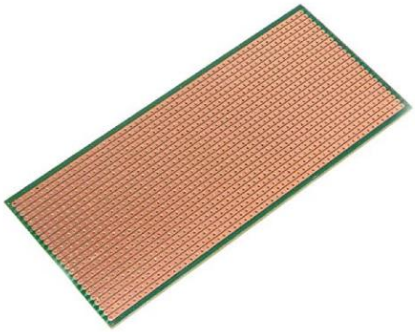
- ✓ Detects accidents through a vibration sensor.
- ✓ Detects alcohol to prevent drunk driving.
- ✓ Sends real-time alerts via Telegram Bot Blynk App integration for remote monitoring and control of the system.

Components List

1. Hardware

 (1) ESP32 microcontroller	 (2) Vibration sensor SM-420
 (3) Alcohol sensor MQ3	 (4) OLED display
 (5) IR sensor	 (6) Buck Converter

 (7) Relay and motor	 (8)Switch

	
(9) Veroboard	(10)Battery

2. Software
- **Blynk App:** For real-time data visualization and control.
- **Telegram Bot:** Sends alert messages with GPS data in case of accidents.

Working Procedure

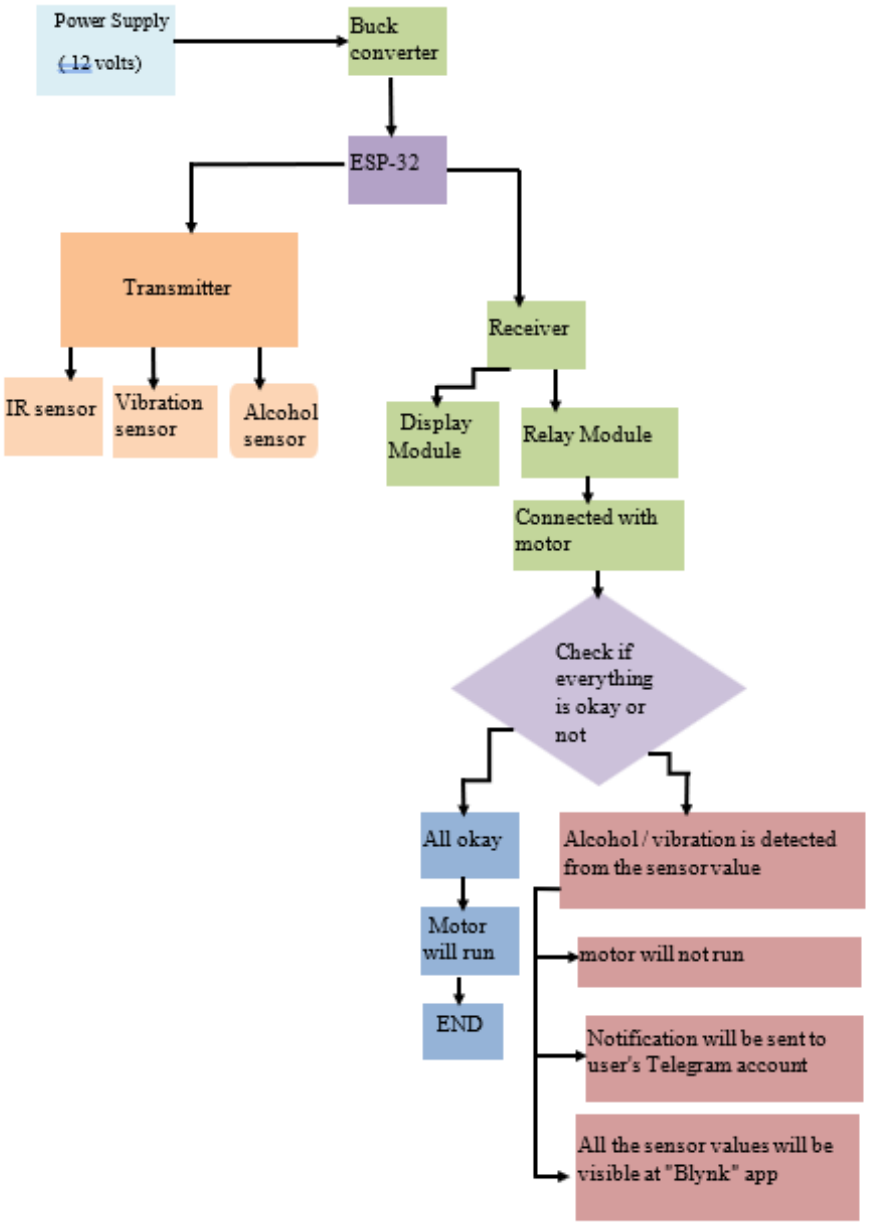
3. The system is powered by a 12V power supply, which is regulated through a buck converter to power the ESP-32 microcontroller.
4. The ESP-32 is connected to a transmitter that collects data from the IR sensor, vibration sensor, and alcohol sensor.
5. Sensor data is transmitted to the ESP-32, which then processes the information.
6. A receiver, display module, and relay module are connected to the ESP-32 to display the sensor values and control the motor.
7. The system checks if the sensor values indicate normal conditions or the presence of alcohol/vibrations.
8. If all values are normal, the motor will run as expected.
9. If the alcohol sensor detects intoxication or the vibration sensor detects an accident, the motor is prevented from starting or is stopped.
10. The system sends a notification to the user's Telegram account, alerting them of the abnormal condition.
11. All sensor data is made visible through the Blynk app, allowing remote monitoring and control.

Github Code Link:

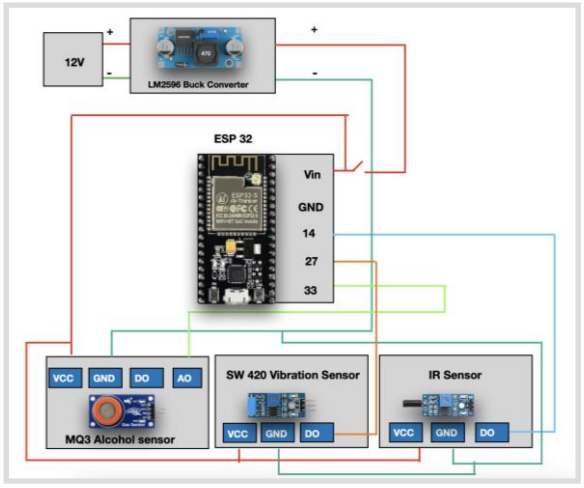
Transmitter

Receiver

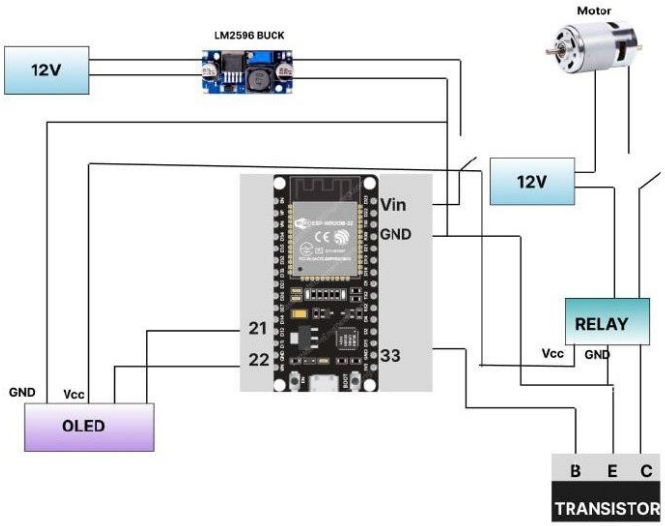
Work Flow


Power Supply
(12 volts)
Buck converter
ESP-32
Transmitter
IR sensor
Vibration sensor
Alcohol sensor
Receiver
Display Module
Relay Module
Connected with motor
Check if everything is okay or not
All okay
Motor will run
END
Alcohol / vibration is detected from the sensor value
motor will not run
Notification will be sent to user's Telegram account
All the sensor values will be visible at "Blynk" app

Transmitter Circuit



Receiver Circuit



Blynk app Integration

The Blynk App is a popular Internet of Things (IoT) platform designed to allow users to control, monitor, and automate IoT devices remotely via a smartphone or web interface. It simplifies the process of building and managing IoT projects by providing a user-friendly interface and a variety of tools to communicate with devices such as ESP32, ESP8266, Raspberry Pi, and Arduino.

In this Accident Detection System, the Blynk app is used to remotely monitor sensor data and control the system in real time. The app provided an intuitive interface for viewing alerts and notifications, such as accident detection or alcohol presence, and allowed seamless communication with the ESP32 through cloud-based connectivity. This enhanced the system's functionality by enabling remote access and control via a smartphone.



Fig: Blynk app interface for alcohol, vibration and IR detection

Cost Analysis

Components	Price (Tk)
ESP32 microcontroller	1000
Vibration sensor SW-420	80
Alcohol sensor MQ3	200
OLED Display	400
IR sensor	70
Buck Converter	160
Relay and Motor	195
Switch	30
Veroboard	80
Battery	330
Total Cost	2545

Future Advancement

1. **Integration with GPS for Precise Location Tracking**: Adding a GPS module would allow the system to send the exact accident location, ensuring quicker response from emergency services.
2. **Machine Learning for Improved Accident Detection**: Using machine learning can help distinguish between minor bumps and real accidents, reducing false alerts.

3. **Expansion to Additional Safety Sensors**: Incorporating smoke or temperature sensors could help detect other emergencies, like vehicle fires, for enhanced safety.
4. **Cloud-based Data Analytics**: Storing accident data in the cloud can help analyze trends over time, offering insights into common causes of accidents.
5. **Voice-Assisted SOS Activation**: A voice recognition feature could allow users to send emergency alerts through simple voice commands when needed.
6. **Integration with Vehicle Diagnostic Systems**: Connecting to vehicle diagnostics can monitor things like brake and engine conditions to prevent accidents before they happen.
7. **Enhanced Mobile Application Features**: Upgrading the app could provide detailed insights, emergency call buttons, and better control through a user-friendly interface.

Conclusion

The Accident Detection System is a practical solution aimed at reducing the risks associated with road accidents and drunk driving. By integrating a vibration sensor for accident detection and an alcohol sensor to prevent impaired driving, the system ensures both proactive and reactive safety measures. Real-time alerts through the Blynk App enable quick responses from emergency contacts, potentially saving lives. With its ease of integration into various vehicle models, this system offers a versatile and effective way to enhance road safety. Future advancements could further improve its functionality, making it an even more comprehensive safety tool for vehicles.